

PRAKTIKUM SISTEM OPERASI

MODUL IV

(MEMBACA SOURCE CODE XINU)



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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JURNAL

1. Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut? Ada pada folder apa file image tersebut?

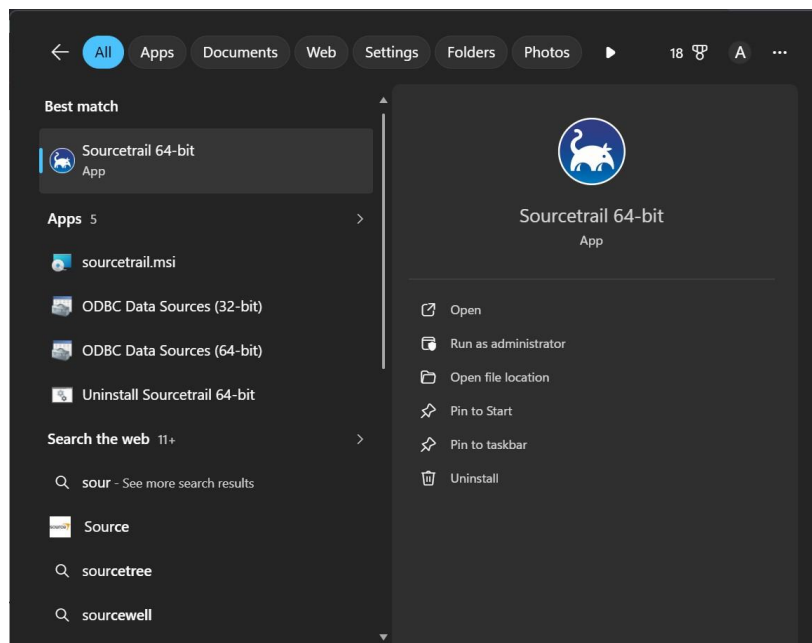
Jawab:

file image hasil kompilasi Xinu bernama xinu.elf, Ukuran file biasanya berkisar antara 500 KB – 2 MB tergantung konfigurasi dan fitur yang dikompilasi, berada di folder ./compile (direktori xinu/compile/)

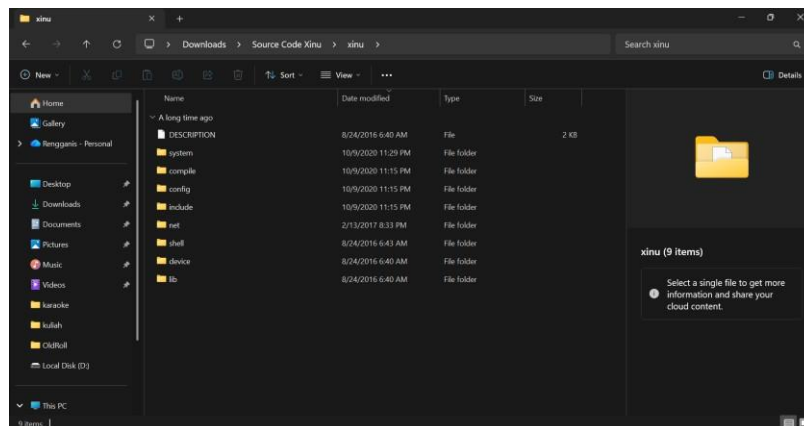
2. Membaca source code Xinu

- a. Download SourceTrail

Jawab:



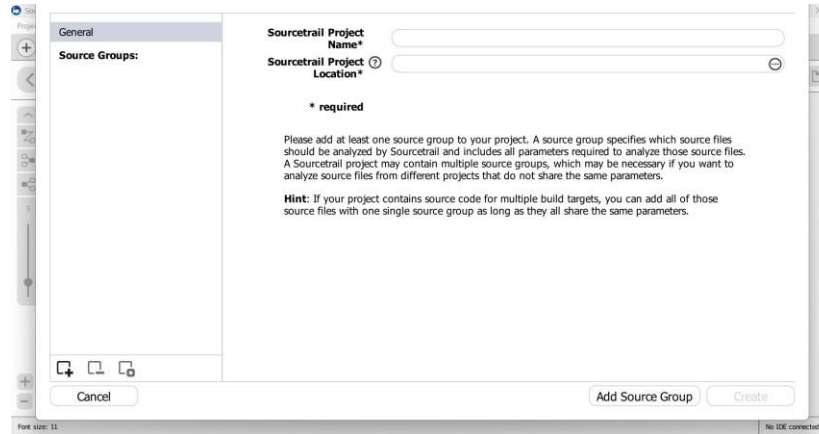
- b. Download file source code xinu



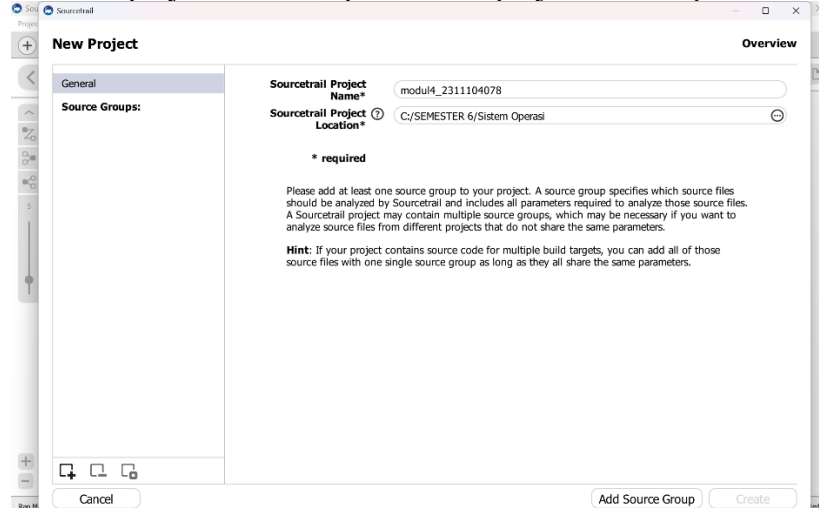
c. Jalankan SourceTrail



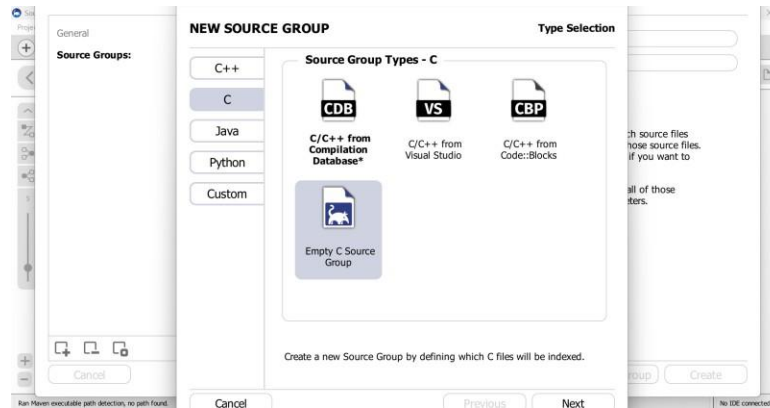
d. Project → New Project



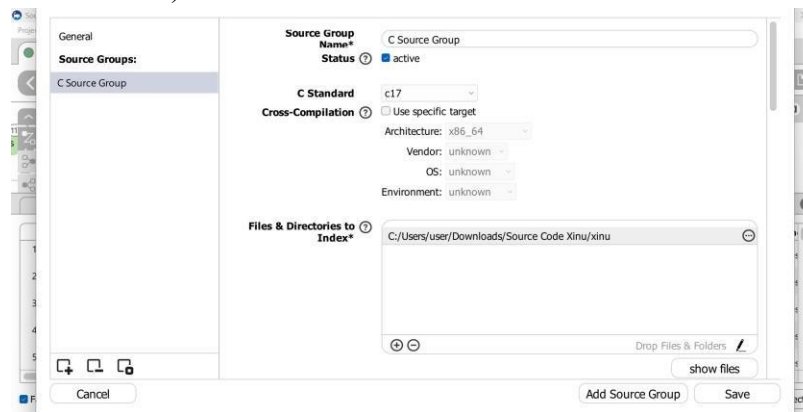
e. Isi nama project xinu dan pilih lokasi project di manapun



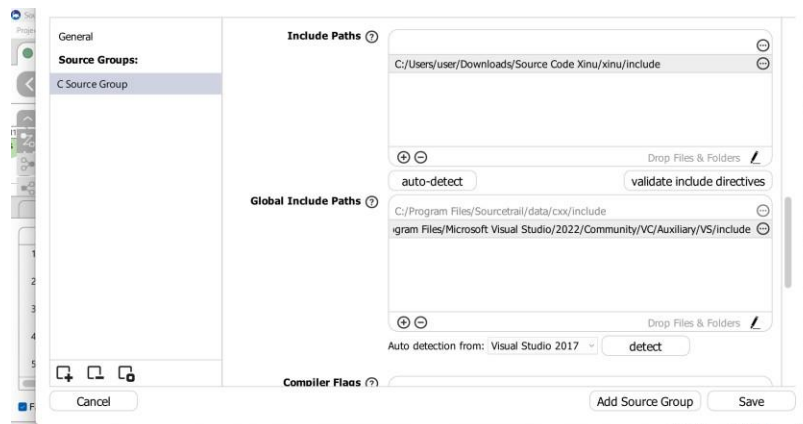
- f. Add Source Groups, pilih C, lalu pilih Empty C Source Group



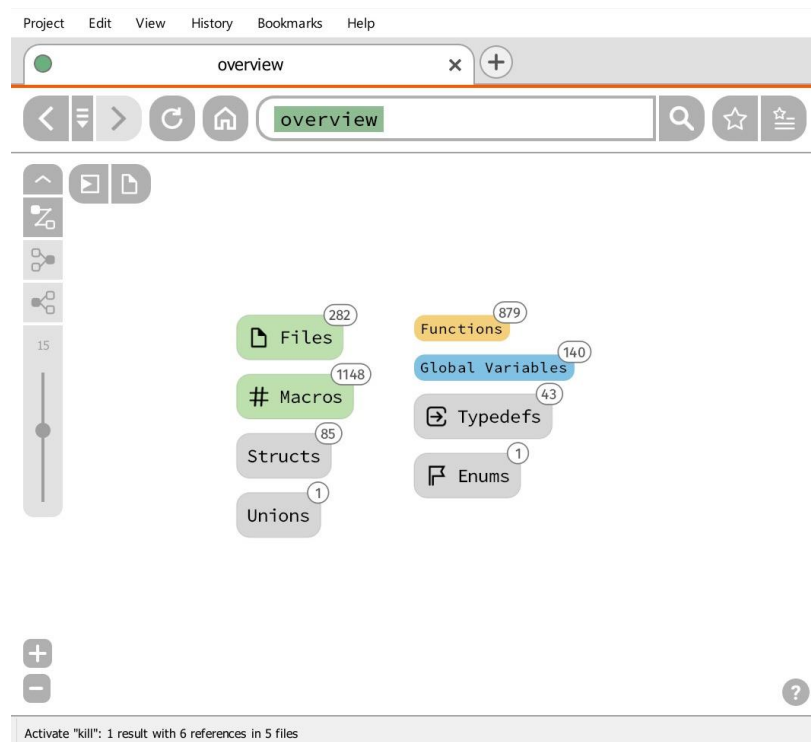
- g. File & Directories to Index: masukkan semua folder Xinu (yang sebelumnya telah di download)



- h. Include Paths: .../xinu/include

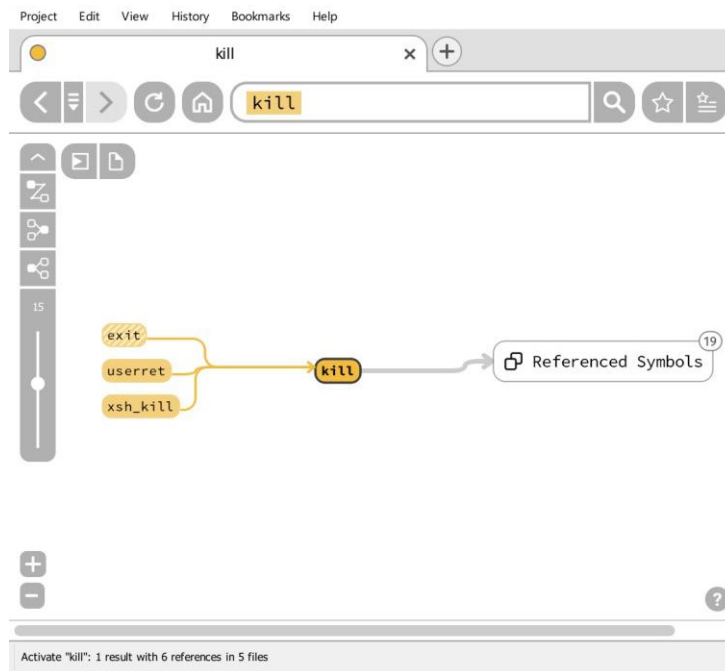


i. Create

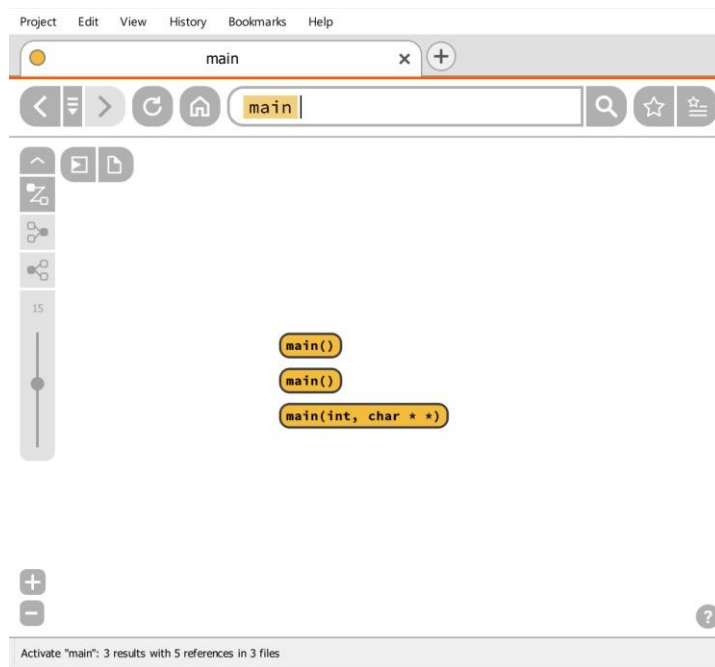


j. Eksplorasi source code Xinu

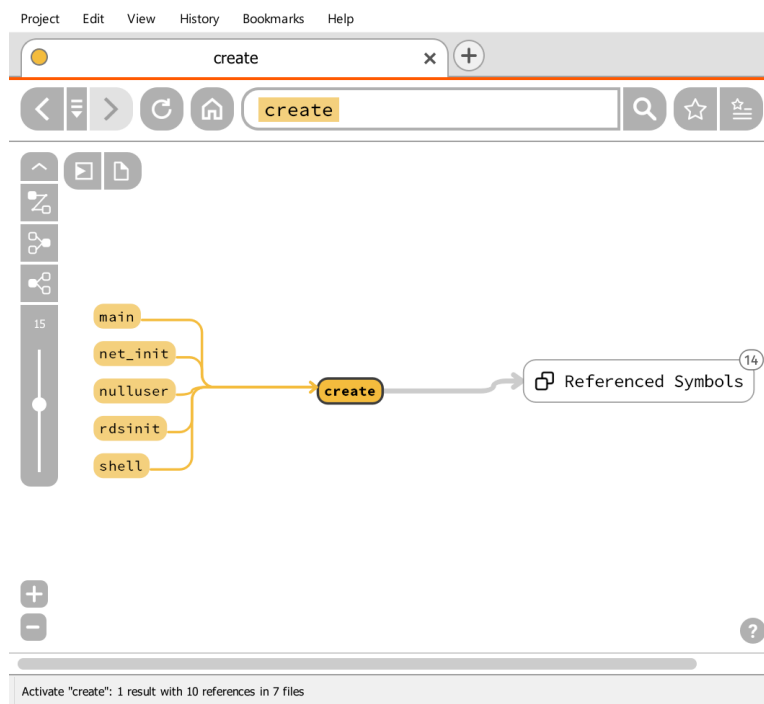
- Kill



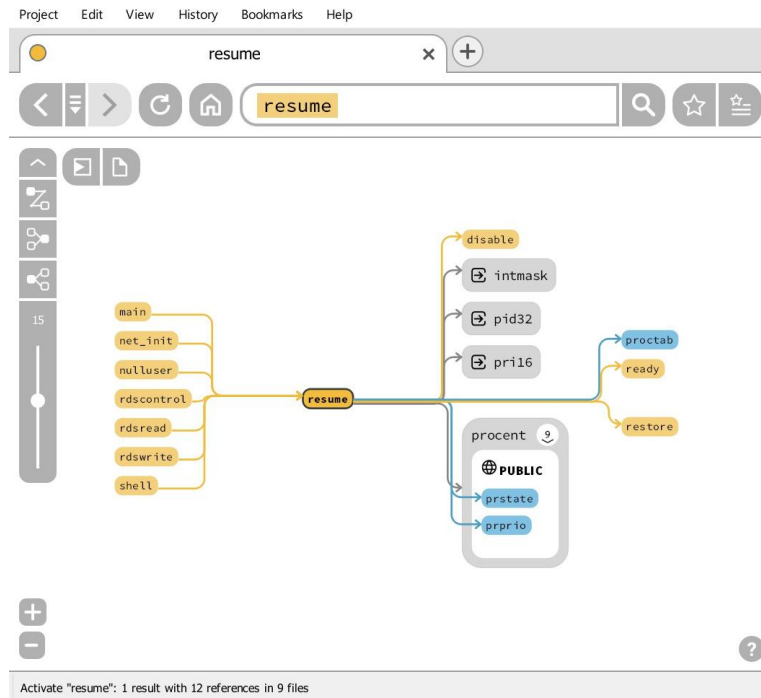
- Main



- Create



- Resume



3. Mencari struktur data dari proses pada Xinu OS. Struktur data proses ada pada file apa? Informasi apa saja yang disimpan dalam struktur data tersebut?

Jawab:

File process.h berlokasi di ~xinu/include/process.h

```

xinu@xinu-develop-end:~$ find / -name "process.h" 2>/dev/null
/home/xinu/prev-version/include/process.h
/home/xinu/xinu/include/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/internal/process.h
/var/lib/dkms/virtualbox-guest/4.3.28/build/include/iprt/process.h
/usr/src/virtualbox-guest-4.3.28/include/internal/process.h
/usr/src/virtualbox-guest-4.3.28/include/iprt/process.h
xinu@xinu-develop-end:~$
  
```

Informasi yang disimpan meliputi

```

xinu@xinu-develop-end:~$ grep -r "procent\|prstate\|prprio\|prstk" /home/xinu/xinu/include/process.h
(proctab[x]).prstate == PR_FREE)
#define NDESC      5      /* Must be odd to make procent 4N bytes */
struct procent {
    uint16 prstate;      /* Process state: PR_CURR, etc. */
    pri16 prprio;        /* Process priority */
    char *prstkptr;      /* Saved stack pointer */
    char *prstkbase;     /* Base of run time stack */
    uint32 prstklen;     /* Stack length in bytes */
extern struct procent proctab[];
xinu@xinu-develop-end:~$ grep -r "struct procent" /home/xinu/xinu/include/process.h
struct procent {      /* Entry in the process table */
xinu@xinu-develop-end:~$ cat /home/xinu/xinu/include/process.h
/* process.h - isbadpid */

/* Maximum number of processes in the system */

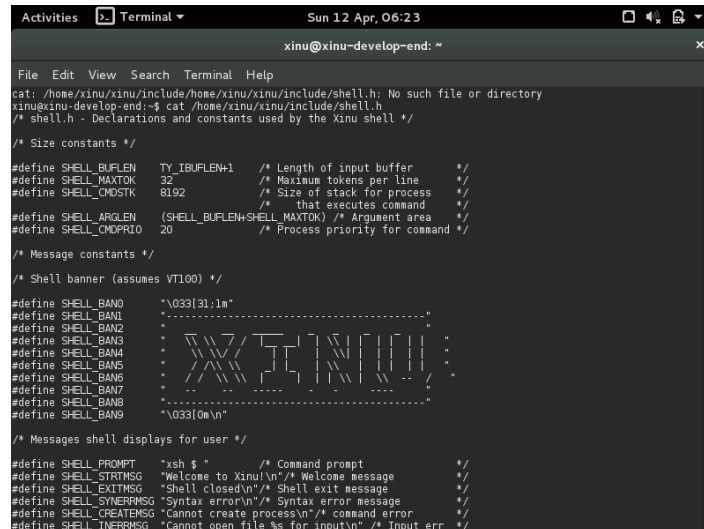
#ifndef NPROC
#define NPROC      8
#endif

/* Process state constants */

#define PR_FREE    0      /* Process table entry is unused */
  
```

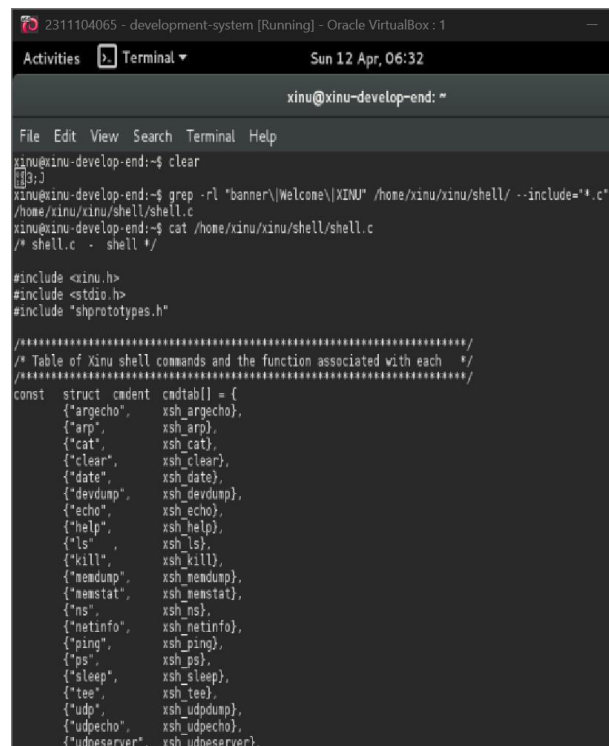
4. Mengubah welcome banner pada Xinu

- a. Carilah file yang menyimpan banner Xinu! Hint: file berekstensi .h pada direktori xinu/include



```
xinu@xinu-develop-end: ~  
File Edit View Search Terminal Help  
cat: /home/xinu/xinu/include/home/xinu/xinu/include/shell.h: No such file or directory  
xinu@xinu-develop-end:~$ cat /home/xinu/xinu/include/shell.h  
/* shell.h - Declarations and constants used by the Xinu shell */  
  
/* Size constants */  
#define SHELL_BUFLen TY_IBUFLen+1 /* Length of input buffer */  
#define SHELL_MaXtOk 32 /* Maximum tokens per line */  
#define SHELL_OmOStK 8192 /* Size of stack for process */  
#define SHELL_ArGLeN (SHELL_BUFLen+SHELL_MaXtOk) /* Argument area */  
#define SHELL_OmPrIO 20 /* Process priority for command */  
  
/* Message constants */  
  
/* Shell banner (assumes VT100) */  
#define SHELL_BaNd "\033[31;1m"  
#define SHELL_BaNd1 "-----"  
#define SHELL_BaNd2 "  
#define SHELL_BaNd3 " XINU / T E M P L A T E "  
#define SHELL_BaNd4 "-----"  
#define SHELL_BaNd5 "  
#define SHELL_BaNd6 " XINU / T E M P L A T E "  
#define SHELL_BaNd7 "-----"  
#define SHELL_BaNd8 "\033[0m\n"  
  
/* Messages shell displays for user */  
  
#define SHELL_PrOmPt "xsh $ " /* Command prompt */  
#define SHELL_STrIMsG "Welcome to Xinu!\n" /* Welcome message */  
#define SHELL_ExITMsG "Shell closed\n" /* Shell exit message */  
#define SHELL_SyNtErRMsG "Syntax error\n" /* Syntax error message */  
#define SHELL_CrEaTEmSg "Cannot create process\n" /* command error */  
#define SHELL_InErRMsG "Cannot open file %s for input\n" /* Input err */
```

- b. Carilah file yang menampilkan banner Xinu! Hint: file berekstensi pada direktori xinu/shell



```
2311104065 - development-system [Running] - Oracle VirtualBox: 1  
Activities Terminal Sun 12 Apr, 06:32  
xinu@xinu-develop-end: ~  
File Edit View Search Terminal Help  
xinu@xinu-develop-end:~$ clear  
[[3;J  
xinu@xinu-develop-end:~$ grep -rI "banner\|Welcome\|XINU" /home/xinu/xinu/shell/ --include="*.c"  
/home/xinu/xinu/shell/shell.c  
xinu@xinu-develop-end:~$ cat /home/xinu/xinu/shell/shell.c  
/* shell.c - shell */  
  
#include <xinu.h>  
#include <stdio.h>  
#include "shprototypes.h"  
  
/* Table of Xinu shell commands and the function associated with each */  
const struct cmdent cmdtab[] = {  
{"argecho", xsh_argecho},  
{"arp", xsh_arp},  
{"cat", xsh_cat},  
{"clear", xsh_clear},  
{"date", xsh_date},  
{"devdump", xsh_devdump},  
{"echo", xsh_echo},  
{"help", xsh_help},  
{"ls", xsh_ls},  
{"kill", xsh_kill},  
{"memdump", xsh_memdump},  
{"memstat", xsh_memstat},  
{"ns", xsh_ns},  
{"netinfo", xsh_netinfo},  
{"ping", xsh_ping},  
{"ps", xsh_ps},  
{"sleep", xsh_sleep},  
{"tee", xsh_tee},  
{"udp", xsh_udpdump},  
{"udpecho", xsh_udpecho},  
{"udpeserver", xsh_udpeserver},  
}
```


- c. Ubahlah welcome banner Xinu (Tambahkan NAMA dan NIM masing-masing, kemudian ubah banner Xinu sesuai dengan selera Anda)

```
/* Shell banner (assumes VT100) */

#define SHELL_BAN0      "\033[31;In"
#define SHELL_BAN1      "-----Naura Aisha Zahira-----"
#define SHELL_BAN2      " "
#define SHELL_BAN3      "  \  \  \ /  _  |  \  \  |  _  |  |  "
#define SHELL_BAN4      "  \  \  \ /  |  |  \  \  |  _  |  |  "
#define SHELL_BAN5      "  /  \  \  |  |  |  \  \  |  _  |  |  "
#define SHELL_BAN6      "  /  \  \  |  |  |  \  \  |  _  |  /  "
#define SHELL_BAN7      "  _  _  _  |  _  _  _  |  _  _  _  |  "
#define SHELL_BAN8      "-----2311104078-----"
#define SHELL_BAN9      "\033[0m\n"
```

Ubah baggian SHELL_BAN0 menjadi nama yaitu Rengganis Tantri Pramudita dan baggian SHELL_BAN8 menjadi NIM yaitu 2311104065

- d. Setelah melakukan perubahan lakukan perintah berikut ini pada terminal Linux
 - \$ cd xinu/compile
 - \$ make clean
 - \$ make
 - \$ sudo minicom
 - Jalankan vm backend

```
-----Naura Aisha Zahira-----
XINU
-----2311104078-----

Welcome to Xinu!
```

